

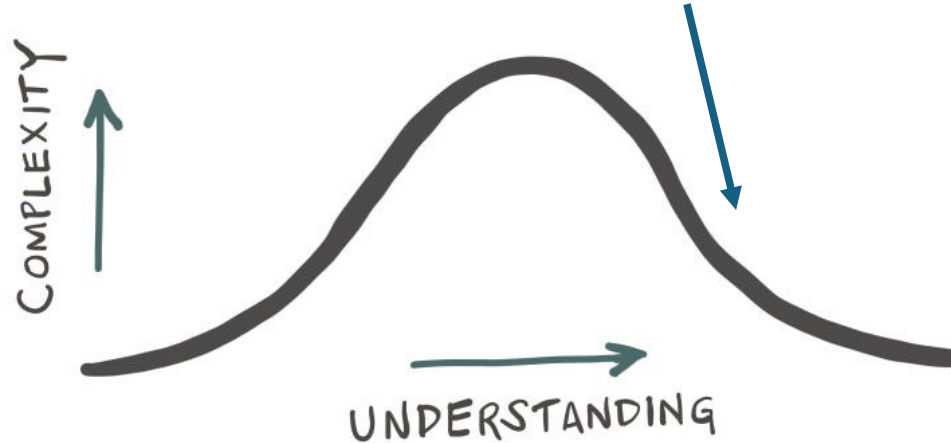
*“as a system evolves, its complexity increases unless work*

*is done to reduce it” — Lehman, 1980 [1]*

**if anyone can do it for \$1 ... can you do it for a cent?**



# easier AI?



[timmm@ieee.org](mailto:timmm@ieee.org)

ssbse'26: july 6, 2026

code: ***pip install ezr***  
data: ***http://tiny.cc/moot***



Ssbse'16

[2]

## An (Accidental) Exploration of Alternatives to Evolutionary Algorithms for SBSE

Vivek Nair<sup>(✉)</sup>, Tim Menzies, and Jianfeng Chen

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vivekax1@gmail.com

**Abstract.** SBSE researchers often use an evolutionary algorithm to solve various software engineering problems. This paper explores an alternate approach of sampling. This approach is called SWAY (Sampling WAY) and finds the (near) optimal solutions to the problem by (i) creating a larger initial population and (ii) intelligently *sampling* the solution space to find the best subspace. Unlike evolutionary algorithms, SWAY does not use mutation or cross-over or multi-generational reasoning to find interesting subspaces but relies on the underlying dimensions of the solution space. Experiments with Software Engineering (SE) models shows that SWAY's performance improvement is competitive with standard MOEAs while, terminating over an order of magnitude faster.

**Keywords:** Search-based SE · Sampling · Evolutionary algorithms



- 2019:  
**Amritanshu Agrawal** (Snap);  
**Vivek Nair** (Meta)  
**Jianfeng Chen** (Meta)
- 2025:  
**Andre Lustosa** (Redhat)
- 2027:  
**Kishan Ganguly** BINGO (FSE'26);  
**Amirali Rayegon** Explanation (JSS'26)
- 2028:  
**Srinath Srinivasan**, Neurosymbolic (arxiv)

**why easier AI?**

how to do easier AI?

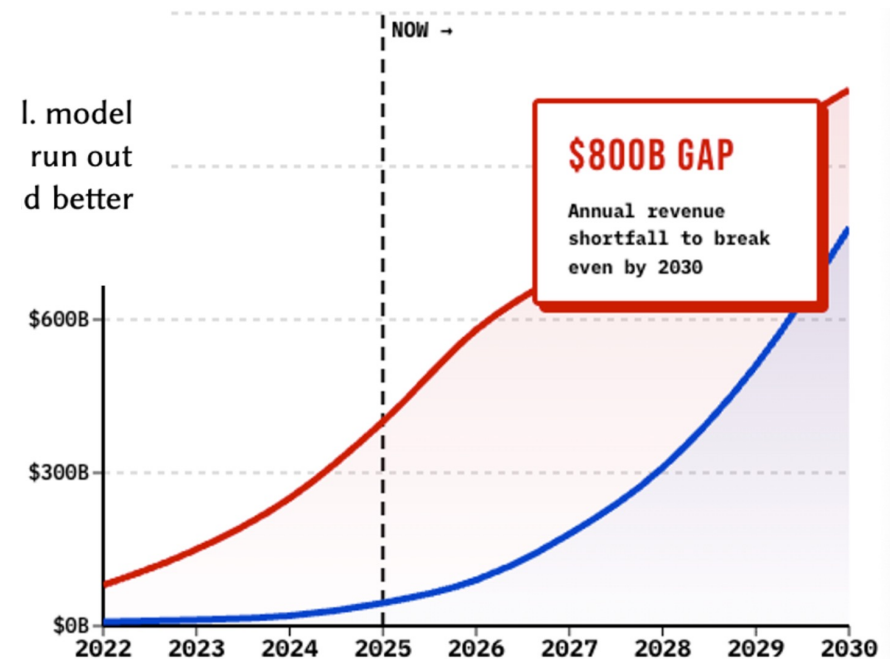
why not more of easier AI?



# Q: why easier AI?

## A: economics [3] [4]

- Niklaus Wirth:
  - “Software gets **slower** faster than hardware gets faster”
- Even if \$/token falls, Jevons paradox: total token burn grows (\*10, \*100: 2023→2026)
- Bain 2025 forecast:
  - \$800B/yr revenue gap by 2030
  - 2026 capex ~\$725B; 2027 projected >\$1T — gap widening



Q: why easier AI?

A: easier to trust; easier to watch and understand



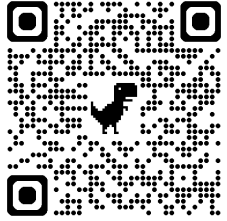
## My Tesla Was Driving Itself Perfectly —Until It Crashed

The danger of almost-perfect tech

By Raffi Krikorian



THE **ATLANTIC**  
MARCH 17, 2026

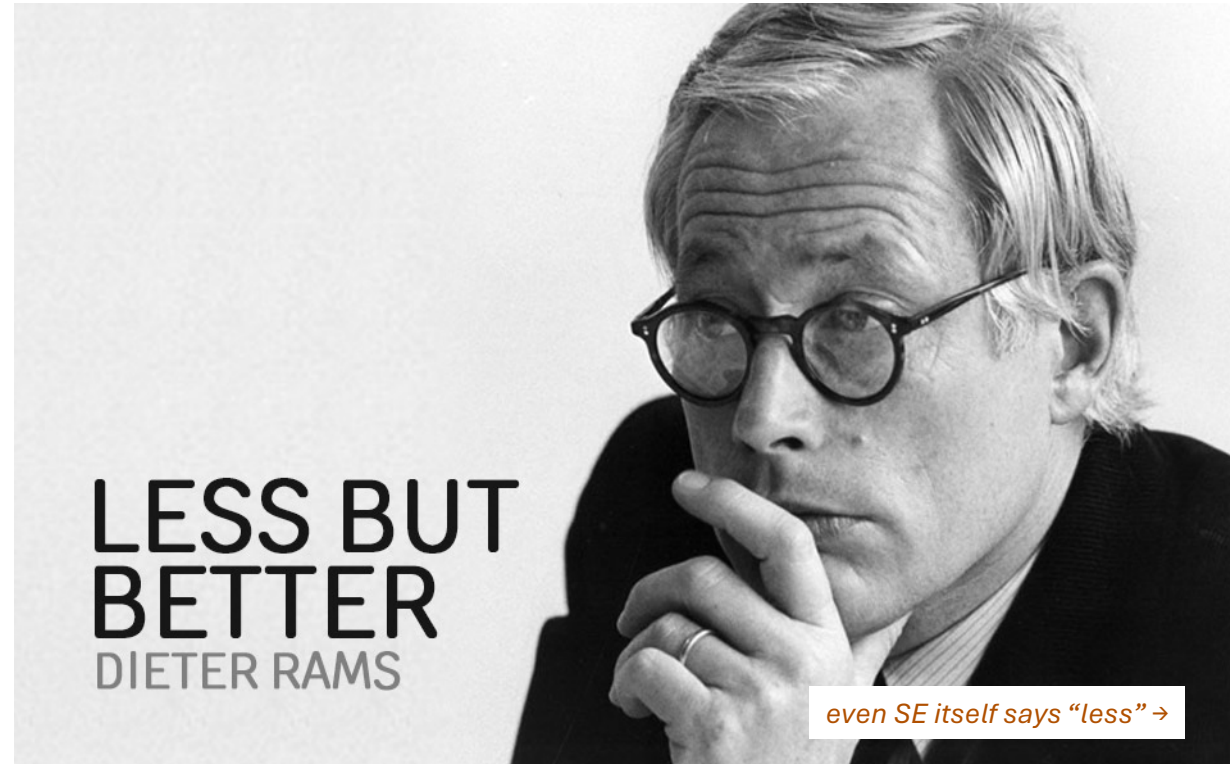


Q: why easier AI?

A: cause some domains demand it [5]

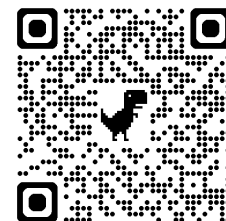
- Green AI
- global **south** (countries without e.g. American-scale infrastructure)
- resource-constraint teaching
  - community colleges,
  - K-12 CS,
  - MSIs,
  - these using **personal** hardware
- privacy-constrained industrial dev
  - **defense**,
  - regulated healthcare,
  - financial services,
  - law firms)
- air-gapped settings;
- **edge** or on-device inference

6

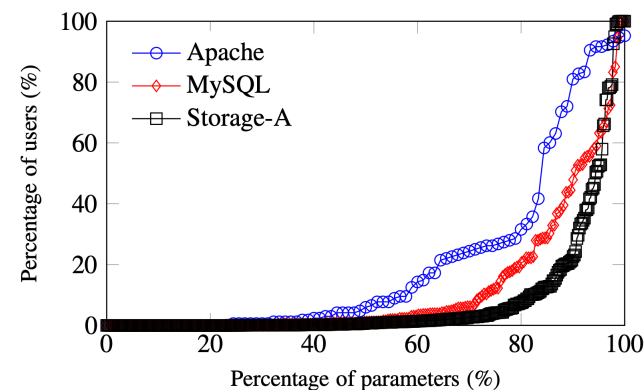
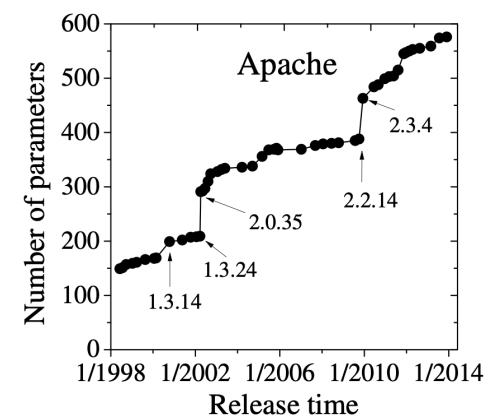


# Q: why easier AI?

## A: software engineering



- **security:**
  - modern software = assembly of pre-made parts (> 80%)
  - more reuse == more exposure
  - e.g. left-pad; MOVEit (\$12B), Falcon (\$5B), ByBit (1.4B) \$24B (2025);
- **survivability:** e.g. SQLite does not use Berkeley DB.
  - Now used on Mars
- **maintainability + reliability:**
  - more loc = more to maintain, more to go wrong
- **waste:** Most users cannot handle most options [6]



and the data for “big” is running out →

Q: why easier AI?

A: cause “big AI” needs too much data [7,10]

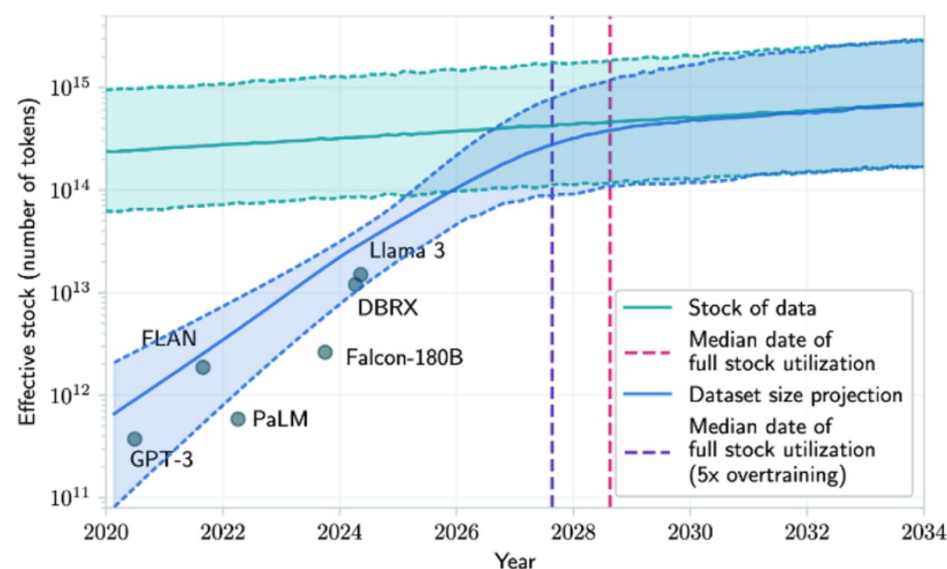
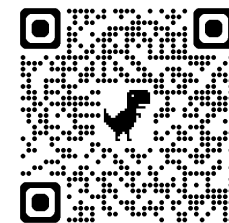
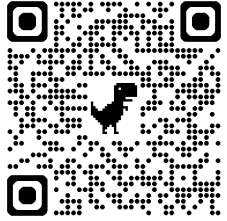


Fig. 4. Median results from the Villalobos et al. model (shown in green) estimate that by 2028, we will run out of new textual data needed to train bigger and better LLMs [75].

*besides, more was never always better →*



Q: why easier AI?

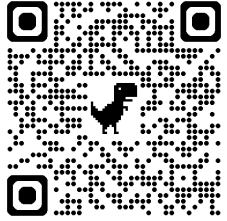
A: more AI not always better [8][9][10]

- text mining predicting effort within **agile** projects,
  - (SVM and TFIDF) better predictions than deep learners
  - Runs 100 times faster
- tree-based models (Random Forest, XGBoost)
  - consistently **outperform** deep neural networks on tabular datasets,
  - especially those with more than 10,000 rows
- Etc [11-14]

why easier AI?

**how to do easier AI?**

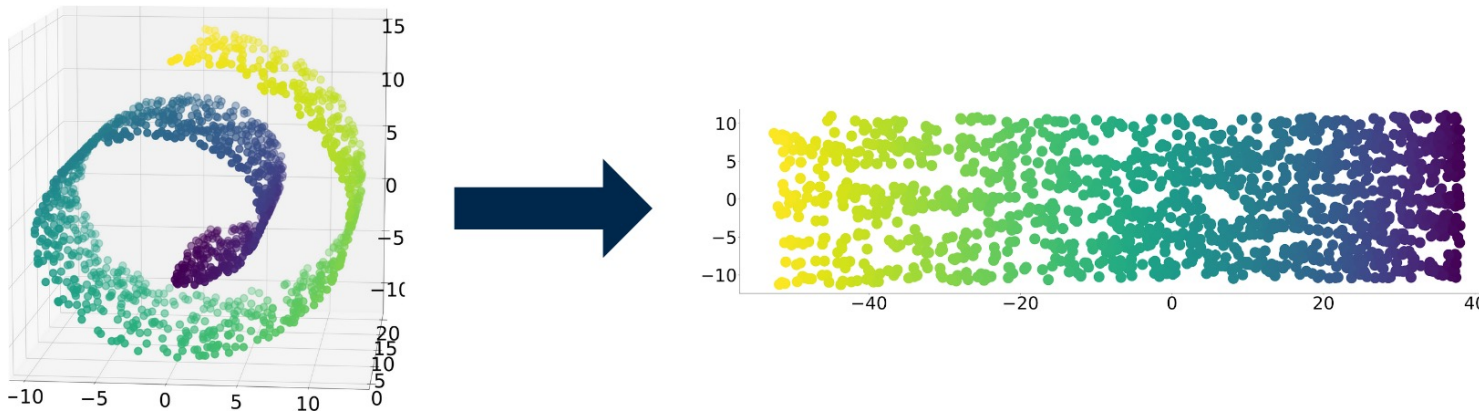
why not more of easier AI?

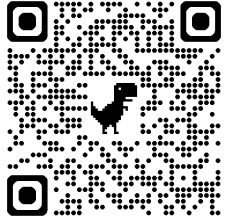


# Q: how to do easier AI?

## A: so many ways [15-28]

- 1300: William of Ockham: “entities must not be multiplied beyond necessity”
- 1902, **PCA**: reduce data to a few principal components
- 1974, **Prototypes**: speed up k-NN by reducing rows to a few exemplars
- 2009, **Active learning**: only use most informative rows
- 2020s, Distillation: compress large LLM models with little performance loss
- etc

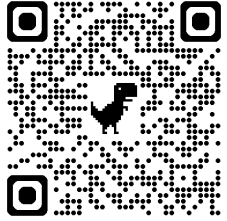




Q: how to do easier AI (with active learning)

A: use current model to decide what's next [29]

- E.g. **SMAC3**
  - ensemble of decision trees
  - Seek unlabeled examples where
    - ensemble most disagrees (a.k.a. explore)
    - **ensemble** most agree (a.k.a. exploit)
    - adapt =  
Slowly shift explore to exploit
- Cost of SMAC3:
  - when new data arrives
  - **rebuilding** the ensemble
- EZR: **pip install ezr**
- “ensemble” = 2 **class** Bayes classifier
  - $|Best|, |Rest| = \sqrt{N}, N - \sqrt{N}$ ,
  - **next** =
    - closest to *Best*
    - furthest from *Rest*
  - update:
    - add next to *Best*,
    - resort *Best*,
    - **pop** worst *Best* to *Rest*
  - Very fast
    - never sort *Rest*
    - Updates to *Best* and *Rest* are fast.



Q: how to test EZR?

A: read lots of papers, collect their data [30]

- Recent SBSE **papers**: ICSE, FSE, TSE, IST, EMSE, TOSEM, ASE
- 127 data sets
- 3-1044 independent **x attributes**;
- 1 to 8 dependent **y attributes** to be minimized or maximized;
- 93 to 100,000 rows.
- <http://tiny.cc/moot>

| #                                                        | Category      | Focus    | File Name(s)     | Here, optimizers struggle with issues like...       | #y  | #x     | Rows     | Refs |
|----------------------------------------------------------|---------------|----------|------------------|-----------------------------------------------------|-----|--------|----------|------|
| <b>Software Systems &amp; Configuration &amp; Tuning</b> |               |          |                  |                                                     |     |        |          |      |
| 24                                                       | Soft. Config  | PLE      | SS-[A-X]         | Minimize footprint/memory vs. maximizing throughput | 2-3 | 3-88   | 197-86k  | [1]  |
| 12                                                       | PromiseTune   | Perf.    | 7z, LLVM, BDBC   | Minimize execution time and energy consumption      | 1   | 6-35   | 864-166k | [2]  |
| 1                                                        | Cloud compute | Tuning   | Apache_AllMeas   | Balance server response vs. CPU/RAM load patterns   | 1   | 9      | 192      | [3]  |
| 1                                                        | Cloud compute | Tuning   | SQL_AllMeas      | Minimize latency/IO vs. maximizing throughput       | 1   | 39     | 4,654    | [4]  |
| 1                                                        | Cloud compute | Tuning   | X264_AllMeas     | Optimize encoding parameters for PSNR/SSIM quality  | 1   | 16     | 1,153    | [1]  |
| 2                                                        | Testing       | Testing  | test120, 600     | Maximize coverage while minimizing execution time   | 1   | 9      | 5,161    | -    |
| <b>Project Management &amp; Process Modeling</b>         |               |          |                  |                                                     |     |        |          |      |
| 35                                                       | Proj. Health  | Health   | Health-Closed    | Optimize PR rates and minimize developer churn      | 2-3 | 5      | 10,001   | [5]  |
| 3                                                        | Scrum         | Agile    | Scrum[1k-100k]   | Maximize velocity within sprint constraints         | 3   | 124    | 1k-100k  | [6]  |
| 8                                                        | Feature Mod.  | Config   | FFM, FM-*        | Optimize Clause/Constraint ratio in large spaces    | 3   | 128-1k | 10,001   | [1]  |
| 1                                                        | nasa93dem     | Cost     | nasa93dem        | Minimize effort (person-months) vs. quality         | 3   | 26     | 93       | [5]  |
| 1                                                        | COC1000       | Risk     | coc1000          | Minimize risks/schedule slip vs. analyst expertise  | 5   | 20     | 1,001    | [7]  |
| 4                                                        | POM3          | Process  | pom3[a-d]        | Balance project idle rates vs. completion costs     | 3   | 9      | 501-20k  | [6]  |
| 4                                                        | XOMO          | Defects  | xomo_[flt,grd]   | Minimize defect density vs. total project effort    | 4   | 27     | 10,001   | [7]  |
| <b>Interdisciplinary &amp; Behavioral Benchmarks</b>     |               |          |                  |                                                     |     |        |          |      |
| 4                                                        | Behavioral    | Behavior | all_players, etc | Maximize user retention and predict cohort churn    | 1-3 | 26-55  | 82-17k   | [8]  |
| 4                                                        | Financial     | Finance  | BankChurners     | Minimize credit risk vs. optimal pricing models     | 2-5 | 19-77  | 1k-20k   | [9]  |
| 3                                                        | Health        | Health   | COVID19, Life    | Maximize accuracy/life vs. readmission likelihood   | 1-3 | 20-64  | 2k-25k   | [10] |
| 2                                                        | RL            | Control  | A2C_[Acr,Crt]    | Maximize reward signals vs. steps to convergence    | 3-4 | 9-11   | 224-318  | -    |
| 5                                                        | Sales         | Market   | Marketing, socks | Maximize ROI vs. minimizing forecasting error       | 1-8 | 20-31  | 247-2k   | [11] |
| 3                                                        | Misc.         | General  | auto93, Car      | Optimize multivariate trade-offs (e.g. MPG vs. HP)  | 2-5 | 5-38   | 205-1k   | [1]  |

configuration, performance tuning, product lines, project health, defect prediction, testing, cost estimation, cross-domain generalization, and text mining

... and a score →



Q: how to score EZR?

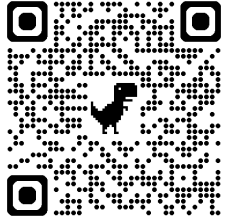
A: win = 1 - normalize(regret) [31]

- Distance to **heaven** (aggregate n goals to one)

$$d2h(y) = \sqrt{\sum_{i=1}^k y_i^2 / \sqrt{k}},$$

- Regret = (**d2h** - lo) / (mu - lo)
- Win = int(100 \* (1 - regret))
  - Win=0 → optimizer just finds mu
    - Failure**
  - Win=100 → found reference optimum
    - Success
- Build regression tree on **labeled** eggs
  - Splitting rows to minimize  $\sigma$  of win
  - Reports what takes you towards good or bad

| win   | n  | Lbs-   | Acc+ | Mpg+ |               |
|-------|----|--------|------|------|---------------|
| 8     | 43 | 2334.2 | 16.3 | 27.4 |               |
| 41    | 26 | 2038.7 | 17.1 | 30.0 | Volume <= 111 |
| 71    | 10 | 1871.7 | 18.1 | 32.0 | Volume <= 83  |
| ✓ 85  | 4  | 1831.7 | 19.0 | 32.5 | Model <= 73   |
| 62    | 6  | 1898.3 | 17.4 | 31.7 | Model > 73    |
| 22    | 16 | 2143.0 | 16.5 | 28.7 | Volume > 83   |
| 36    | 7  | 2098.1 | 18.0 | 28.6 | Model <= 73   |
| 12    | 9  | 2177.9 | 15.3 | 28.9 | Model > 73    |
| 26    | 4  | 2210.0 | 15.7 | 30.0 | Volume > 97   |
| 0     | 5  | 2152.2 | 15.0 | 28.0 | Volume <= 97  |
| -41   | 17 | 2786.1 | 15.1 | 23.5 | Volume > 111  |
| -9    | 5  | 2279.8 | 14.6 | 28.0 | Volume <= 116 |
| -54   | 12 | 2997.1 | 15.3 | 21.7 | Volume > 116  |
| -44   | 8  | 2921.1 | 15.8 | 22.5 | Model > 73    |
| -38   | 4  | 2563.0 | 14.0 | 25.0 | Clntrs <= 4   |
| -50   | 4  | 3279.2 | 17.4 | 20.0 | Clntrs > 4    |
| ✗ -75 | 4  | 3149.0 | 14.4 | 20.0 | Model <= 73   |

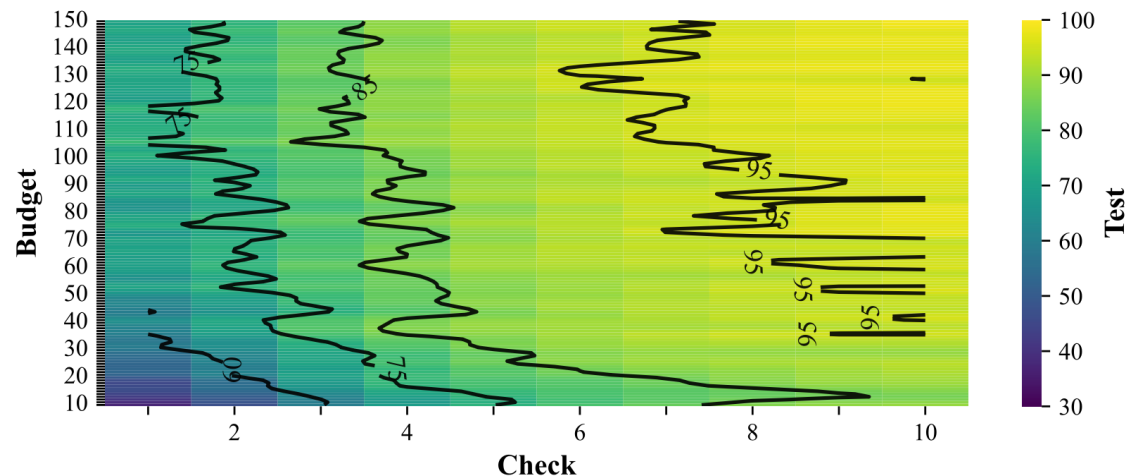


Q: how well does EZR generalize?

A: sort holdout via tree, check first few [31]

$\text{Win} = \text{int}(100 * (1 - \text{regret}))$

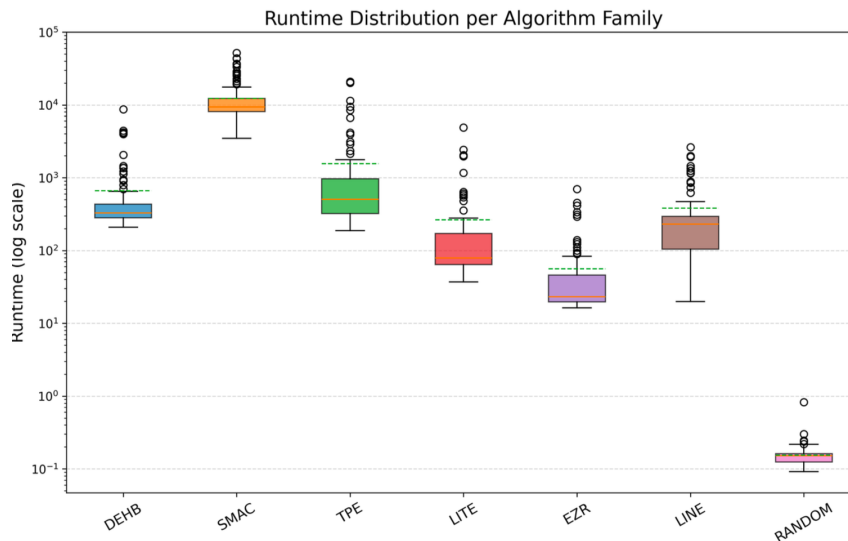
- 100,000 times
  - pick any MOOT data set
  - Train:
    - Active learning, labeling y rows with a **limited** budget.
    - Build tree from labels
  - Test:
    - Use tree to sort holdout
    - **Check** first few
- Active learning, very effective
  - Label 50 rows
  - Check 5
  - **Achieve** 85% of best



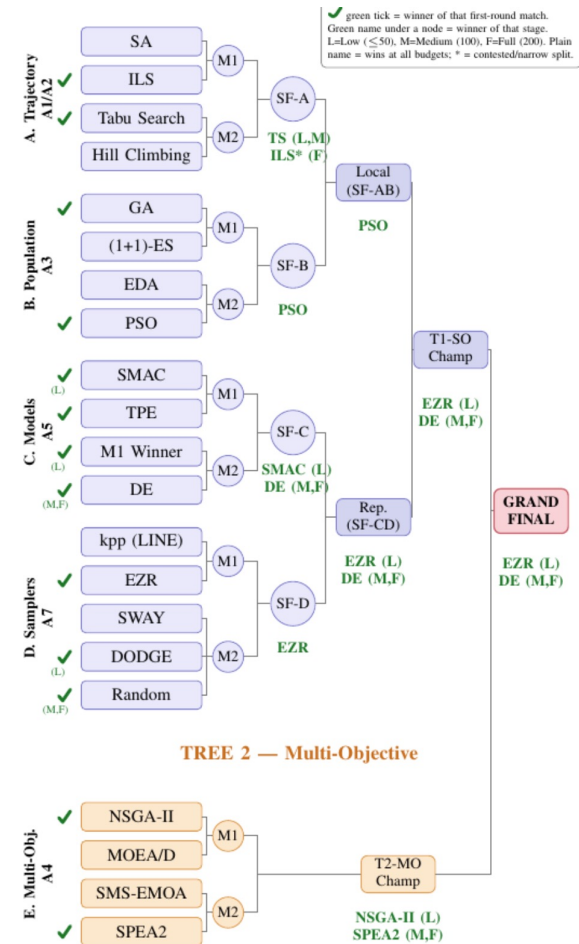
This entire 100,000-experiment study: one laptop, 2 hours, no cluster, no cloud bill.

Q: how fast and effective  
are EZR's trees?

A: very and very [32]



500x less CPU = 500x less energy. every run.

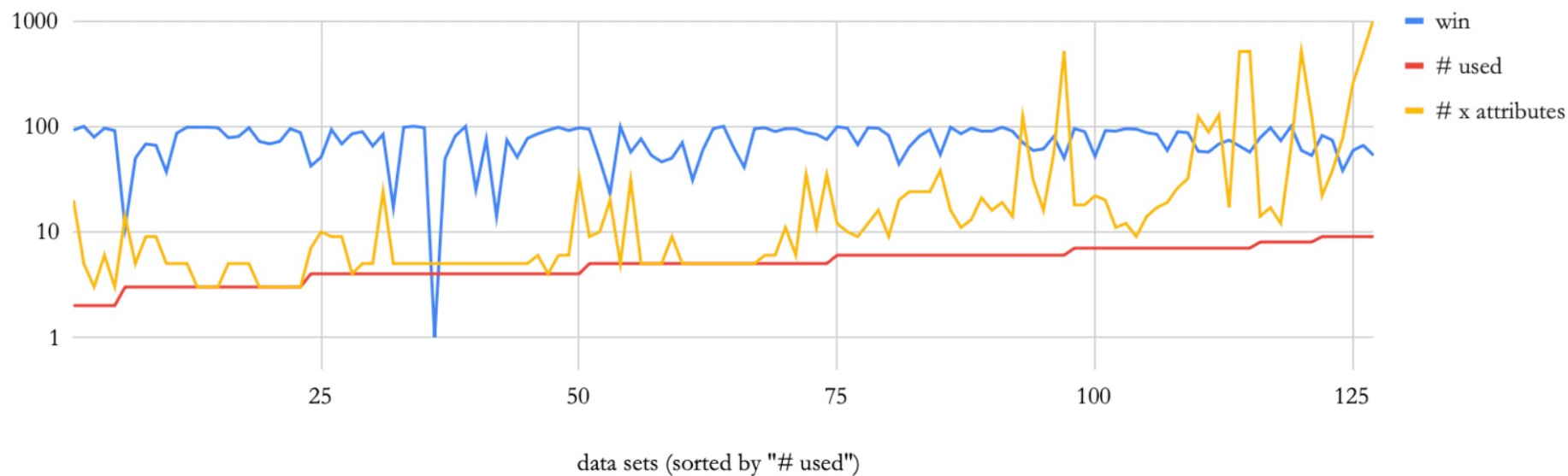
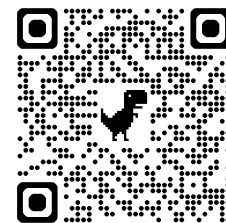


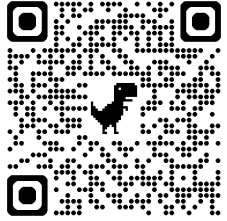
e.g. NSGA-II needs 1000 samples for what EZR reaches in 50

and are the models small enough to explain? →

Q: how big are EZR's models?

A: surprisingly small [33]





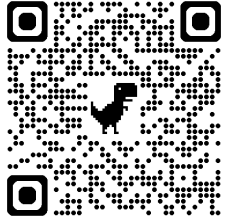
Q: so is EZR useful for explanation?

A: EZR's trees competitive with SOTA [33]

- Build trees using N **attributes**
- N= #attributes used in EZR's tree
- Using other tools: select **top** N attributes
- White = tied best
- Blue = within **top** 90%
- Yellow = within top 75%
- EZR goes a long way
  - E.g. last row
  - For a **3 goal** task
    - EZR uses 7 / 1044 attributes
    - EZR uses 150 rows
    - Arguably competitive
- **across 127 datasets: trees use**
- **4–9 of up to 1044 attributes**

| RLF | SHAP | EZR | anova | all | data        | x* | x    | y | budget | rows   |
|-----|------|-----|-------|-----|-------------|----|------|---|--------|--------|
| 95  | 100  | 100 | 100   | 100 | HSMGP       | 6  | 14   | 1 | 150    | 3457   |
| 87  | 81   | 82  | 82    | 85  | SS-R        | 7  | 14   | 2 | 150    | 3009   |
| 94  | 95   | 63  | 57    | 96  | SS-V        | 4  | 16   | 2 | 150    | 6841   |
| 74  | 74   | 59  | 74    | 99  | SS-W        | 7  | 16   | 2 | 150    | 65537  |
| 100 | 100  | 100 | 100   | 100 | X264-AM     | 6  | 16   | 1 | 150    | 1153   |
| 100 | 100  | 100 | 99    | 100 | SS-M        | 7  | 17   | 3 | 150    | 865    |
| 81  | 81   | 81  | 81    | 81  | SS-N        | 8  | 17   | 2 | 150    | 53663  |
| 59  | 62   | 53  | 52    | 58  | coc1000     | 9  | 20   | 5 | 150    | 1001   |
| 95  | 96   | 96  | 93    | 100 | SS-U        | 7  | 21   | 2 | 150    | 4609   |
| 98  | 98   | 93  | 98    | 98  | xomo_flight | 7  | 27   | 4 | 150    | 10001  |
| 92  | 92   | 90  | 90    | 92  | xomo_grnd   | 7  | 27   | 4 | 150    | 10001  |
| 94  | 95   | 95  | 95    | 95  | xomo_osp    | 7  | 27   | 4 | 150    | 10001  |
| 87  | 88   | 82  | 82    | 88  | xomo_osp2   | 8  | 27   | 4 | 150    | 10001  |
| 77  | 75   | 71  | 64    | 85  | SQL-AM      | 10 | 39   | 1 | 150    | 4654   |
| 87  | 91   | 92  | 74    | 94  | billing10k  | 7  | 88   | 3 | 150    | 10001  |
| 75  | 75   | 81  | 75    | 97  | Scrum100k   | 7  | 124  | 3 | 150    | 100001 |
| 82  | 88   | 92  | 88    | 97  | Scrum10k    | 8  | 124  | 3 | 150    | 10001  |
| 83  | 89   | 86  | 83    | 89  | Scrum1k     | 7  | 124  | 3 | 150    | 1001   |
| 81  | 99   | 93  | 81    | 99  | FFM-125     | 9  | 128  | 3 | 150    | 10001  |
| 81  | 97   | 89  | 97    | 97  | FFM-250     | 8  | 256  | 3 | 150    | 10001  |
| 95  | 93   | 93  | 95    | 95  | FFM-500     | 8  | 510  | 3 | 150    | 10001  |
| 81  | 96   | 87  | 96    | 96  | FM-500-1    | 8  | 511  | 3 | 150    | 10001  |
| 86  | 91   | 93  | 91    | 97  | FM-500-2    | 7  | 511  | 3 | 150    | 10001  |
| 95  | 95   | 95  | 87    | 95  | FM-500-3    | 9  | 513  | 3 | 150    | 10001  |
| 97  | 97   | 93  | 96    | 97  | FM-500-4    | 9  | 517  | 3 | 150    | 10001  |
| 82  | 94   | 94  | 71    | 100 | FFM-1000    | 7  | 1044 | 3 | 150    | 10001  |

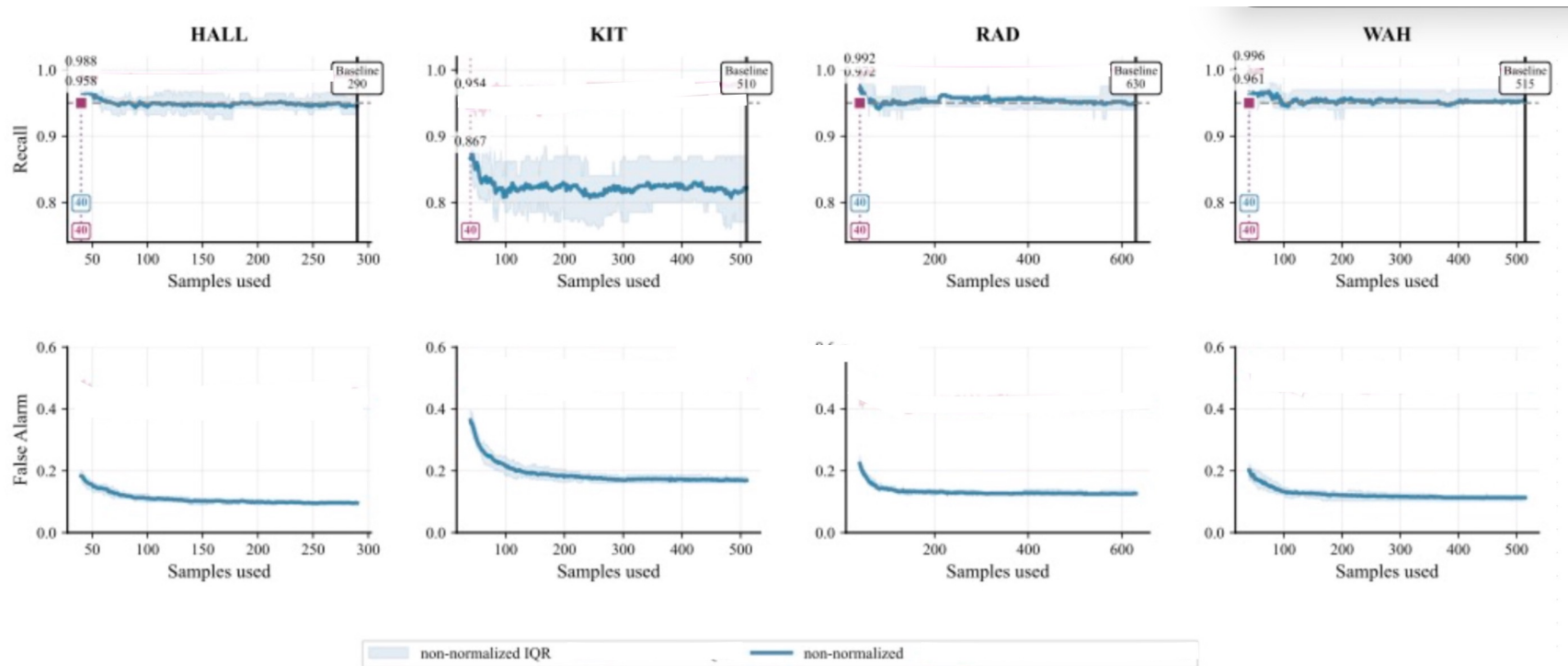
so why isn't everyone doing this? →



Q: what does EZR say about LLMs?

A: LLMs rarely tested against simpler tool [34]

- Search google scholar for **relevant** research



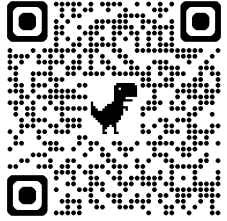
why easier AI?

how to do easier AI?

**why not more of easier AI?**

Q: if so good, why not more easier AI?

A: (weak) empirical standards [34] [35]



- In a survey of 229 SE papers on **LLMs in SE**
  - Only 5% compared to simpler approaches
- **Methodological** error
  - Other methods can produce results that are better and/or **faster**
    - (See above)
- My career:
  - Step 1: compare "it" vs a **stupid** straw man
  - Step 2: throw "it" away, use the straw man



*also, the new toys are seductive →*

Q: if so good, why not more easie  
A: cause “big” sells; prompts/loo

- Welcome to my data center
  - ribbon on the door the mayor can cut
  - **such** a spectacle!
- Compare that to **timm inc.**
  - No special software to sell
    - you can build it yourself
  - No special hardware to sell
    - our algorithms are so fast, they don't **need** it
  - IPO : low evaluation

```
make sh
[nothing] 0:htop*
15 if (s[1:] if s[1:] == "-" else s.isdigit(): return
16 try: return float(s)
17 except ValueError: return s=="True" or (s!="False" a
18
19 def settings(s):
20     "Parse every var=val in a string into an o (vals co
21     pairs = re.findall(r"(\w+)=(\S+)", s)
22     return o(**{k: thing(v) for k, v in pairs})
23
24 def csv(file, clean=lambda s: s.partition("#")[0].spl
25     "Yield typed rows from a CSV file ('#' starts a com
26     with open(file, encoding="utf-8") as f:
27         for line in f:
28             row = [x.strip() for x in clean(line)] # strip
29             if any(row): # skip
30                 yield tuple(thing(x) for x in row) # hash
31
32 def say(x, dec=2):
33     "Pretty string: whole floats as ints, else `dec` pl
34     if isa(x, float):
35         return str(int(x)) if x == int(x) else f"{x:.{dec}
36     if isa(x, o): x = vars(x) # unwrap a name
37     if isa(x, dict): return "{" + ", ".join(f"{k}: {say(
38         for k, v in sorted(x.items(
~/gits/aiez/nuff/nuff.py

Time: 00:00:00.000000 0[|||||
Mem[|||||17.66/24.0G] 1[|||||
Swp[|||||1.40G/3.00G] 3[|||||
F1Help F2Setup F3SearchF4FilterF5Tree F6SortByF7Nice F8
```

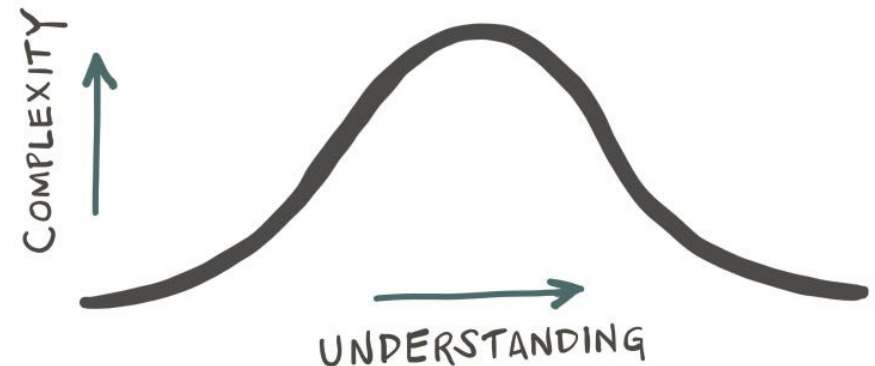


and simplicity is genuinely hard →

Q: if so good, why not easier AI?  
A: the simplicity paradox



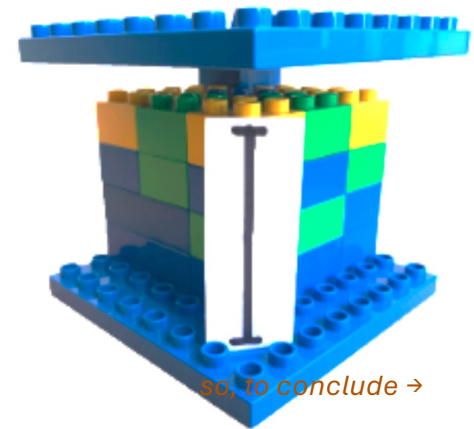
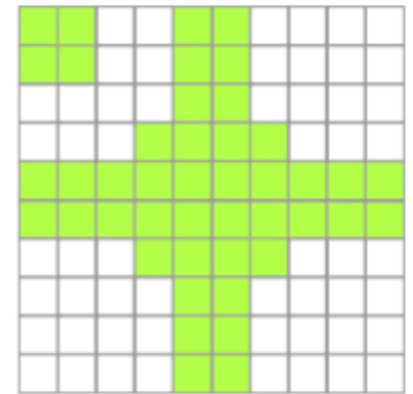
- Simplicity needs **experience**.
  - “Experience is the name every one gives to their **mistakes**.”  
— *Lady Windermere’s Fan* (1892)
- Simplicity is complicated to certify
  - You have to build and/or **run** simpler “it” and complex “it”
  - Needs lot of CPU



Q: if so good, why not more easier AI?  
A: human cognitive bias [37]



- Our simpler methods have not been **previously** explored
  - since other researchers **were not looking for them.**
- Why?
- Humans are biased to bloat
  - In **studies** of 100s of human subjects
    - Humans favor additive to subtractive changes
    - 1,155 to 297 (about 4:1)



**TL;DR**



# Lehman's **2nd** law <sup>[1]</sup>

- “as a system evolves, its complexity **increases** unless work is done to maintain or reduce it”
- Q: why / how can we **reduce** the complexity of AI?
- A: we must / we can
- **do**: baseline the simple:  
    pip install ezr + tiny.cc/moot
- **review**:
  - ask “compared to what simpler?”
- **teach**:
  - easier AI first; scale only when simple fails

/ (question | comment)\* /

code: *pip install ezr*  
data: *<http://tiny.cc/moot>*

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